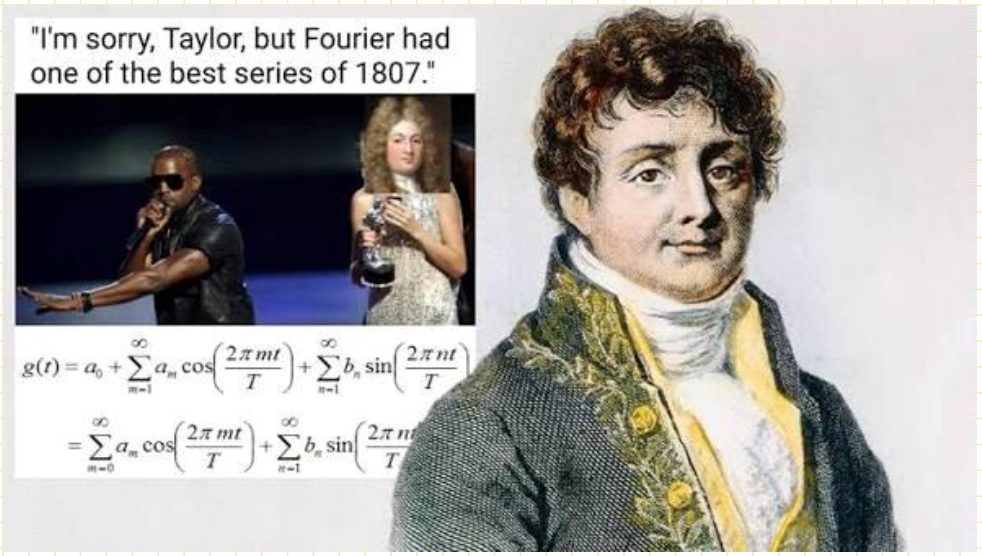
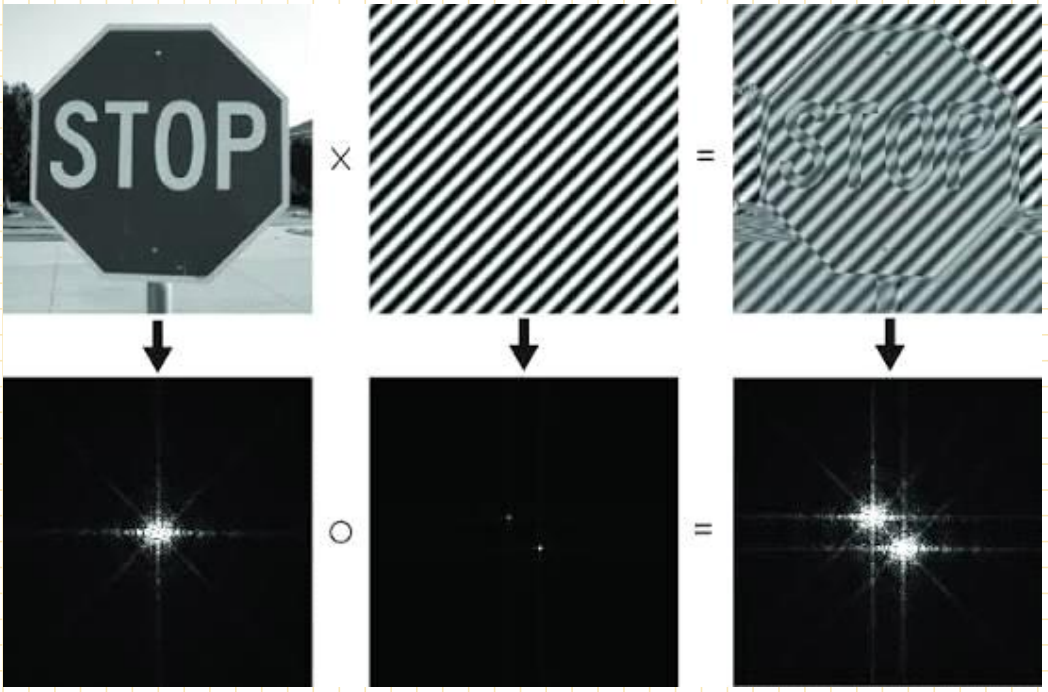
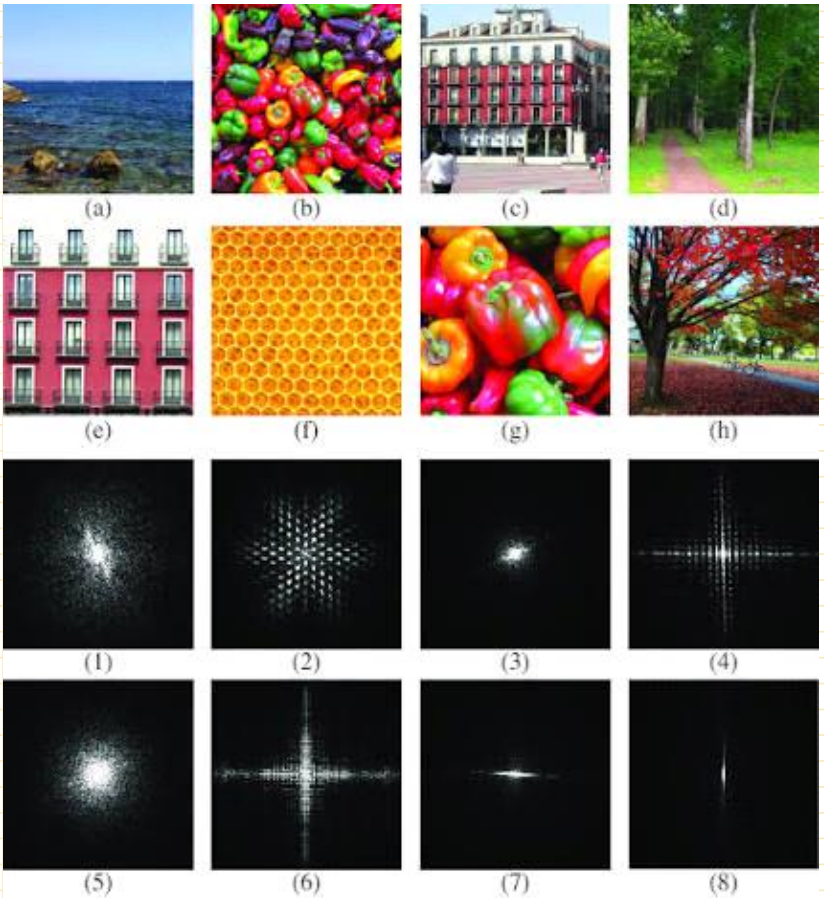
Fourier Transform

Monday, 19 October 2026 10:04 am







FOURIER COSINE and SINE TRANSFORMS

An **integral transform** is a transformation in the form of an integral that produces from given functions new functions depending on a different variable. One is mainly interested in these transforms because they can be used as tools in solving ODEs, PDEs, and integral equations and can often be of help in handling and applying special functions. The Laplace

Fourier Cosine Transform

The Fourier cosine transform concerns ***even functions*** $f(x)$. We obtain it from the Fourier cosine integral [(10) in Sec. 10.7]

$$f(x) = \int_0^\infty A(w) \cos wx \, dw, \qquad \text{where} \qquad A(w) = \frac{2}{\pi} \int_0^\infty f(v) \cos wv \, dv.$$

Namely, we set $A(w) = \sqrt{2/\pi}\, \hat{f}_c(w)$, where c suggests "cosine." Then, writing $v = x$ in the formula for $A(w)$, we have

(1a) $$\hat{f}_c(w) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos wx \, dx$$

and

(1b) $$f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{f}_c(w) \cos wx \, dw.$$

Formula (1a) gives from $f(x)$ a new function $\hat{f}_c(w)$, called the **Fourier cosine transform** of $f(x)$. Formula (1b) gives us back $f(x)$ from $\hat{f}_c(w)$, and we therefore call $f(x)$ the **inverse Fourier cosine transform** of $\hat{f}_c(w)$.

The process of obtaining the transform $\hat{f}_c$ from a given f is also called the **Fourier cosine transform** or the *Fourier cosine transform method.*

Fourier Sine Transform

Similarly, in (11), Sec. 11.7, we set $B(w) = \sqrt{2/\pi}\, \hat{f}_s(w)$, where s suggests "sine." Then, writing $v = x$, we have from (11), Sec. 11.7, the **Fourier sine transform**, of $f(x)$ given by

(2a) $$\hat{f}_s(w) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin wx \, dx,$$

and the **inverse Fourier sine transform** of $\hat{f}_s(w)$, given by

(2b) $$f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{f}_s(w) \sin wx \, dw.$$

The process of obtaining $f_s(w)$ from $f(x)$ is also called the **Fourier sine transform** or the *Fourier sine transform method.*

Other notations are

$$\mathcal{F}_c(f) = \hat{f}_c, \qquad \mathcal{F}_s(f) = \hat{f}_s$$

and $\mathcal{F}_c^{-1}$ and $\mathcal{F}_s^{-1}$ for the inverses of $\mathcal{F}_c$ and $\mathcal{F}_s$, respectively.

EXAMPLE 1 **Fourier Cosine and Fourier Sine Transforms**

Find the Fourier cosine and Fourier sine transforms of the function

$$f(x) = \begin{cases} k & \text{if} \quad 0 < x < a \\ 0 & \text{if} \quad x > a \end{cases} \qquad \text{(Fig. 285).}$$

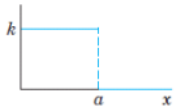


Fig. 285. $f(x)$ in Example 1

Solution. From the definitions (1a) and (2a) we obtain by integration

$$\hat{f}_c(w) = \sqrt{\frac{2}{\pi}}\, k \int_0^a \cos wx\, dx = \sqrt{\frac{2}{\pi}}\, k \left(\frac{\sin aw}{w}\right)$$

$$\hat{f}_s(w) = \sqrt{\frac{2}{\pi}}\, k \int_0^a \sin wx\, dx = \sqrt{\frac{2}{\pi}}\, k \left(\frac{1 - \cos aw}{w}\right).$$

This agrees with formulas 1 in the first two tables in Sec. 11.10 (where $k = 1$).
Note that for $f(x) = k = \text{const}$ $(0 < x < \infty)$, these transforms do not exist. (Why?)

EXAMPLE 2 **Fourier Cosine Transform of the Exponential Function**

Find $\mathcal{F}_c(e^{-x})$.

Solution. By integration by parts and recursion,

$$\mathcal{F}_c(e^{-x}) = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-x} \cos wx\, dx = \sqrt{\frac{2}{\pi}} \frac{e^{-x}}{1 + w^2} (-\cos wx + w \sin wx) \Big|_0^\infty = \frac{\sqrt{2/\pi}}{1 + w^2}.$$

This agrees with formula 3 in Table I, Sec. 11.10, with $a = 1$. See also the next example.

PROPERTIES OF FOURIER COSINE & SINE TRANSFORMS

1 - Linearity: $\mathcal{F}_c[af+bg] = a\mathcal{F}_c[f]+b\mathcal{F}_c[g]$

$\mathcal{F}_s[af+bg] = a\mathcal{F}_s[f]+b\mathcal{F}_s[g]$,

where f and g are functions, $a, b \in \mathbb{R}$.

[Comment: $\hat{f}_c(w) = \mathcal{F}_c[f]$, $\hat{f}_s(w) = \mathcal{F}_s[f]$.]

Proof: Exercise.

$$\mathcal{F}_c[af+bg] = \sqrt{\frac{2}{\pi}} \int_0^\infty (af(x)+bg(x)) \cos\omega x \, dx.$$

$$= a\sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos\omega x \, dx + a\sqrt{\frac{2}{\pi}} \int_0^\infty g(x) \cos\omega x \, dx$$

$$= a\,\mathcal{F}_c[f] + b\,\mathcal{F}_c[g].$$

Similarly, for $\mathcal{F}_s$.

2 - Fourier Sine & Cosine Transforms of Derivatives:

Let $f(x)$ be continuous and absolutely integrable on the x-axis, let $f'(x)$ be piecewise continuous on each finite interval, and let $f(x) \to 0$ as $x \to \infty$.

Then,

$$\mathcal{F}_c[f'] = \omega\,\mathcal{F}_s[f] - \sqrt{\frac{2}{\pi}} f(0).$$

$$\mathcal{F}_s[f'] = -\omega\,\mathcal{F}_c[f]$$

Proof: Exercise.

$$\mathcal{F}_c[f'] = \sqrt{\frac{2}{\pi}} \int_0^\infty f'(x) \cos\omega x \, dx = \sqrt{\frac{2}{\pi}} \left[f(x) \cos\omega x \right]_0^\infty + \omega \int_0^\infty f(x) \sin\omega x \, dx$$

$$= -\sqrt{\frac{2}{\pi}} f(0) + \omega\,\mathcal{F}_s[f].$$

Similarly,

$$\mathcal{F}_s[f'] = \sqrt{\frac{2}{\pi}}\int_0^{\infty} f'(x)\sin\omega x\,dx = \sqrt{\frac{2}{\pi}}\Big[f(x)\sin\omega x\Big]_0^{\infty} - \sqrt{\frac{2}{\pi}}\int_0^{\infty} f(x)\cos\omega x\,dx$$

$$= 0 - \omega\,\mathcal{F}_c[f] = -\omega\,\mathcal{F}_c[f].$$

Hence,

$$\mathcal{F}_c[f'] = \omega\,\mathcal{F}_s[f] - \sqrt{\frac{2}{\pi}}f(0).$$

$$\mathcal{F}_s[f'] = -\omega\,\mathcal{F}_c[f].$$

Exercise: Show that

(a) $\mathcal{F}_c[f''] = -\omega^2\mathcal{F}_c[f] - \sqrt{\frac{2}{\pi}}f'(0).$

(b) $\mathcal{F}_s[f''] = -\omega^2\mathcal{F}_s[f] + \sqrt{\frac{2}{\pi}}f(0).$

EXAMPLE 3 **An Application of the Operational Formula (5)**

Find the Fourier cosine transform $\mathcal{F}_c(e^{-ax})$ of $f(x) = e^{-ax}$, where $a > 0$.

Solution. By differentiation, $(e^{-ax})'' = a^2e^{-ax}$; thus

$$a^2f(x) = f''(x).$$

From this, (5a), and the linearity (3a),

$$a^2\mathcal{F}_c(f) = \mathcal{F}_c(f'')$$

$$= -w^2\mathcal{F}_c(f) - \sqrt{\frac{2}{\pi}}f'(0)$$

$$= -w^2\mathcal{F}_c(f) + a\sqrt{\frac{2}{\pi}}.$$

Hence

$$(a^2 + w^2)\mathcal{F}_c(f) = a\sqrt{2/\pi}.$$

The *answer* is (see Table I, Sec. 11.10)

$$\mathcal{F}_c(e^{-ax}) = \sqrt{\frac{2}{\pi}}\left(\frac{a}{a^2 + w^2}\right) \qquad (a > 0).$$

Exercise :

1) Find the Fourier Sine & Cosine Transform of $f(x)=\begin{cases} k, & 0<x<a \\ 0, & x>a. \end{cases}$

2) Find $\mathcal{F}_c[e^{-x}]$ and $\mathcal{F}_c[e^{-ax}]$ where $a>0$.

3) $\mathcal{F}_s[2e^{-3x}+3e^{-2x}]$

4) Show that Fourier Sine & Cosine Transforms of x^{n-1} are $\frac{\Gamma(n)}{\omega^n}\sin\frac{n\pi}{2}$ and $\frac{\Gamma(n)}{\omega^n}\cos\frac{n\pi}{2}$, respectively, and where $\Gamma(n)$ = Gamma Function of $n = \int_0^\infty e^{-t}t^{n-1}\,dt$.

5) Find Fourier Sine & Cosine Transform of $f(x)=e^{-x}$, and hence show that $\int_0^\infty \frac{\cos mx}{1+x^2}\,dx=\frac{\pi}{2}e^{-m}$ and $\int_0^\infty \frac{x\sin mx}{1+x^2}\,dx=\frac{\pi}{2}e^{-m}$.

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1. Find the cosine transform $\hat{f}_c(w)$ of $f(x) = 1$ if $0 < x < 1$, $f(x) = -1$ if $1 < x < 2$, $f(x) = 0$ if $x > 2$.

9. Find $\mathcal{F}_s(e^{-ax})$, $a > 0$, by integration.

11. Find $f_s(w)$ for $f(x) = x^2$ if $0 < x < 1$, $f(x) = 0$ if $x > 1$.

12. Find $\mathcal{F}_s(xe^{-x^2/2})$ from (4b) and a suitable formula in Table I of Sec. 11.10.